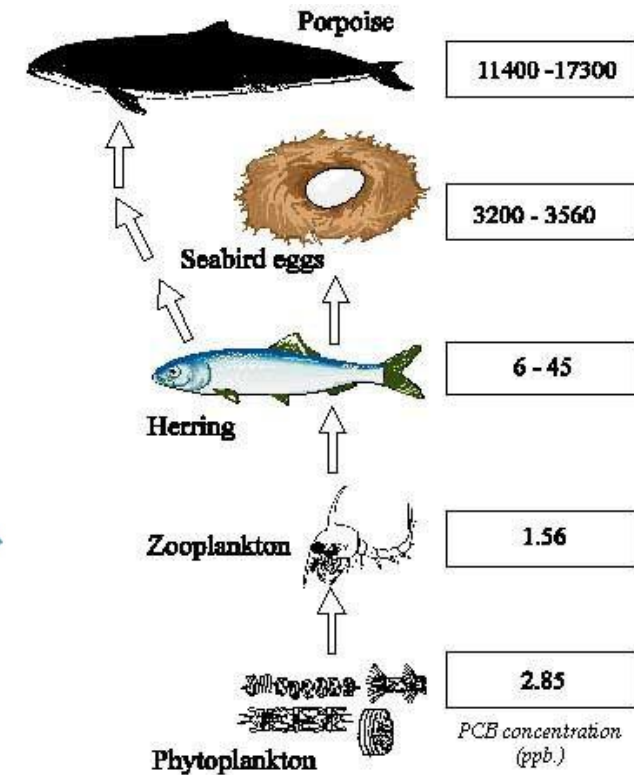
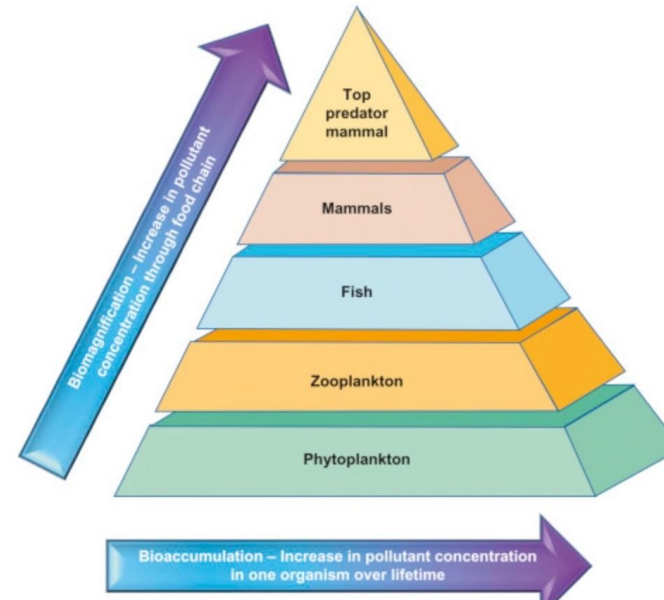
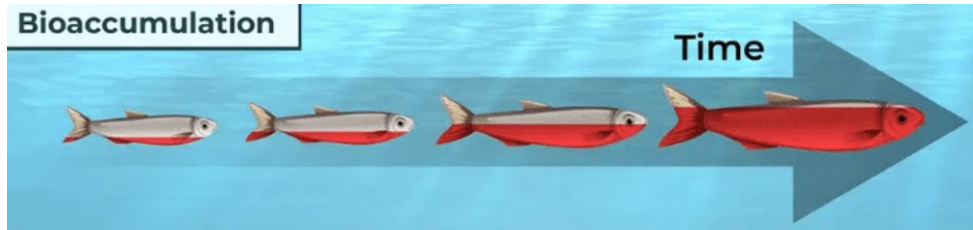


- **Bioaccumulation** and **biomagnification** are two different processes that often occur in tandem with one another.
- Bioaccumulation is the process by which toxins enter the food web by building up in **individual organisms**
- Biomagnification is the process by which toxins are passed from one trophic level to the next (and thereby increase in concentration) within a food web.



The concentration of PCB's tends to increase in the tissues of organisms at higher levels in the marine food chain, a phenomenon termed 'biomagnification'.
[From: Percy, Wells and Evans MS, 1996: see Further Reading]

An Introduction to Bioremediation

“**Bios**” means life and refers to living organisms

“**To remediate**” that means to solve a problem.

- “Bioremediate” means to use biological organisms to solve an environmental problem such as contaminated soil or groundwater.
- It is a technology for removing pollutants from the environment thus restoring the original natural surroundings and preventing further pollution

Advantages of bioremediation

- Completely natural process with almost no harmful side effects
- Can be done *in-situ* for most applications with no dangerous transport
- Lowers liability, as contaminants are less likely to escape
- Little energy consumed compared to incineration and landfilling
- High acceptance from regulatory authorities
- Cost effective

Limitations:

- Range of contaminants that can be effectively treated **is limited** to compounds that are biodegradable.
- In some cases, microbial metabolism of contaminants **may produce toxic metabolites**.
- The process is sensitive to the level of toxicity and environmental conditions in the ground i.e. the conditions must be conducive to microbial activity
- Treatment time is typically longer than that of other remediation technologies.

- Is the contaminant biodegradable?
- Is biodegradation occurring in the site naturally?
- Are environmental conditions appropriate for biodegradation?
- If the waste does not completely biodegrade, where will it go?

Types of Bioremediation

On the basis of place where wastes are removed, there are principally two ways of bioremediation

In Situ Bioremediation

- Most often, in situ bioremediation is applied to eliminate the pollutants in contaminated soils and groundwater.
- It is a superior method for the cleaning of contaminated environments because it saves transportation costs and uses harmless microorganisms to eliminate the chemical contaminations.
- Also, the method is preferred as it causes the least disruption of the contaminated area. This would be of much relevance either where the least investment and pollution are favored (for example in factories) or in areas contaminated with dangerous contaminants (for example in areas contaminated with chemical or radioactive materials)

Biostimulation

- The stimulation of the indigenous microbial populations in soils and/or ground water.
- Biostimulation process is mediated through application of nutrient supplements, substrate molecules for microbial growth, and introduction of microbial strains having higher capability of degradation

Alteration or manipulation of environmental conditions to enhance microbial growth & activity



Lack of proper condition to survive & growth

Bio-stimulation



Addition of adequate nutrients (N, P, S, Ca, Mg etc)

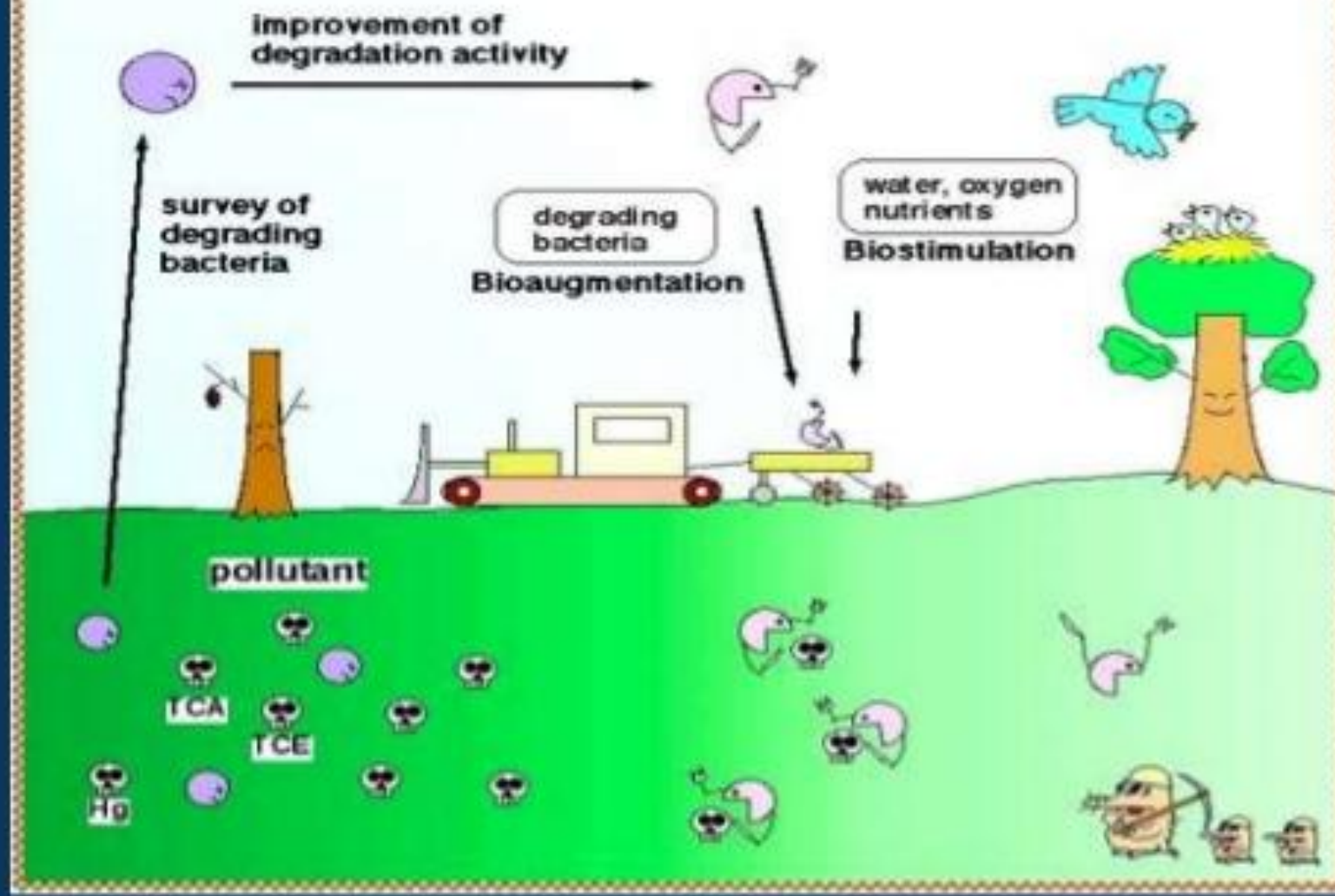
Bioaugmentation

- This type of bioremediation is performed through the **introduction of certain microorganisms** to a contamination site.
- As the conditions of contamination sites are most often unfavorable for the establishment and bioactivity of the exogenously amended microorganisms, therefore here like intrinsic bioremediation, Oxygen, and nutrients (for example nitrogen and phosphorus) are required to enhance microbial growth

Disadvantages of in situ bioremediation

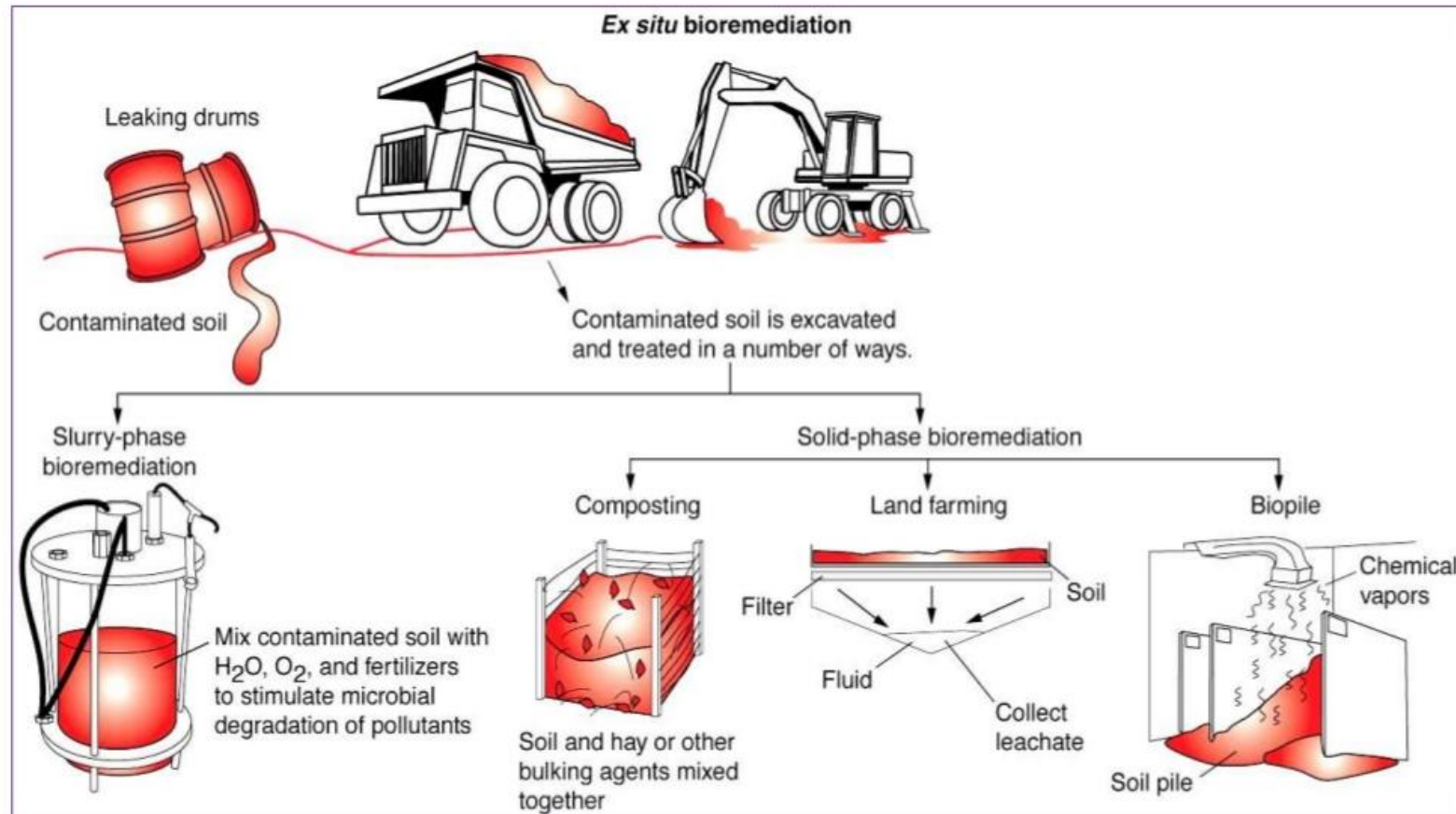
- Very time consuming process
- Sites are directly exposed to environmental factors such as temperature, oxygen supply etc.
- Microbial degrading ability varies seasonally

Bioremediation

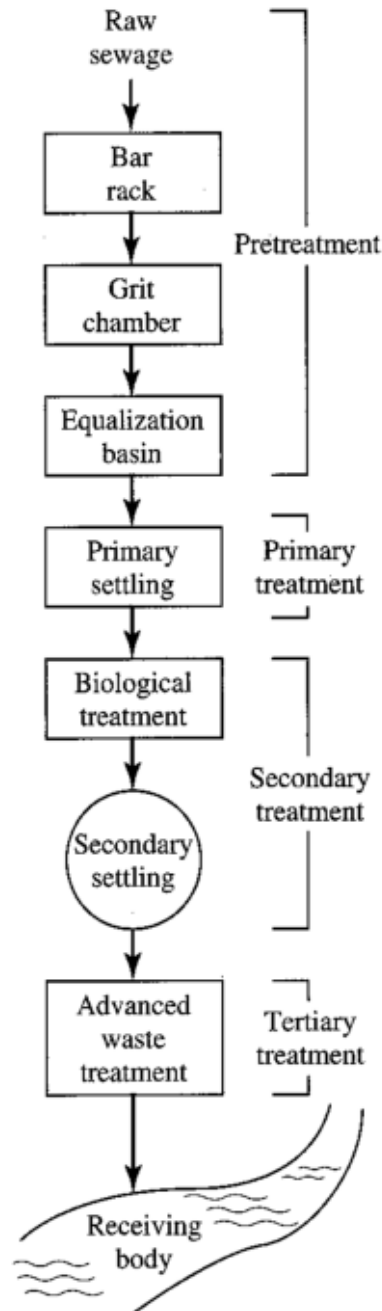


Ex Situ Bioremediation

- The process of bioremediation here takes place **somewhere out from contamination site**, and therefore requires **transportation** of contaminated soil or pumping of groundwater to the site of bioremediation.
- This technique has more disadvantages than advantages.



Municipal Wastewater Treatment Systems



- **Preliminary treatment** (removes materials that can cause operational problems, equalization basins are optional)
- **Primary treatment** (remove ~60% of solids and ~35% of BOD)
- **Secondary treatment** (remove ~85% of BOD and solids)
- **Advanced treatment** (varies: 95+ % of BOD and solids, N, P)
- **Final Treatment** (disinfection)
- **Solids Processing** (sludge management)

Preliminary of Industrial Wastewaters

Industrial wastewaters must be pretreated prior to being discharged to municipal sewer system

- Pretreatment requirements set by regulatory agencies
- **Why:** remove materials that will not be treated by municipal system, remove materials that inhibit the biological processes in secondary treatment
- **For example:** silver ions are toxic to bacteria which might affect biological process. Thus silver ions are removed at pre-treatment before biological process so that bacterial performance do not get affected
- Preliminary treatment **serves to prepare wastewater for subsequent treatment, but it effects little change in wastewater quality.**

Primary treatment

- Primary treatment is the second step in wastewater treatment.
- It allows for the physical separation of solids and greases from the wastewater - usually involves gravity sedimentation of screened, degrittied wastewater to remove settleable solids
- The screened wastewater flows into a primary settling tank where it is held for several hours allowing solid particles to settle to the bottom of the tank and oils and greases to float to the top
- Slightly more than **one-half of the suspended solids (60%)** ordinarily are removed.
- BOD in the form of solids removable by sedimentation **(30%)** is also removed.
- Primary treatment effects some reduction in the effluent concentration of nutrients, pathogenic organisms, trace elements, and potentially toxic organic compounds.
- The constituents that are removed are contained in **primary sludge**

Secondary treatment

- biological treatment process** that removes dissolved organic material from wastewater. The partially treated wastewater from the settling tank flows by gravity into an aeration tank.
- mixing of water to solids containing that use oxygen to consume the remaining organic matter in the wastewater as their food supply (use of air bubble for mixing and oxygen supply)
- liquid mixture (i.e., solids with micro-organisms and water) is sent to the final clarifier.
- In clarifier, solids settle out to the bottom where some of the material is sent to the solids handling process and some is recycled back to replenish the population of micro-organisms in the aeration tank to treat incoming wastewater.

Final treatment

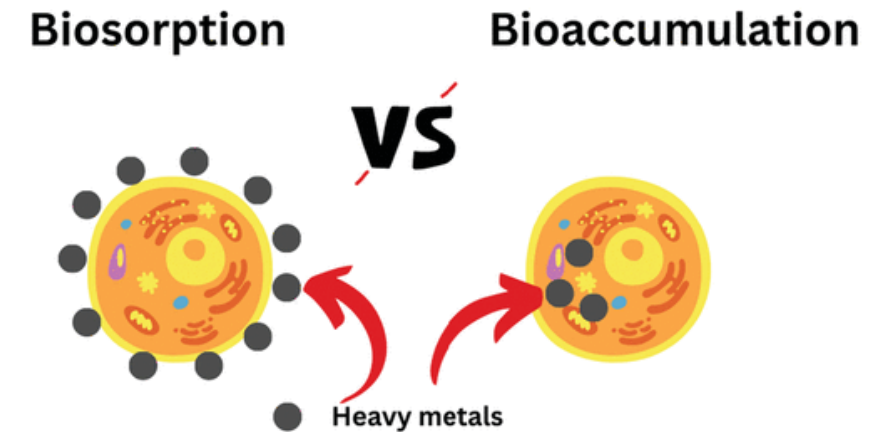
- Treated water is disinfected and then it is send out for wastewater reuse activities or for discharging in river/streams.
- Disinfection for control of pathogenic microorganisms and viruses is the most common type of tertiary treatment.
- **Mostly chlorination and/or ultra violet irradiation is used for disinfection purposes.**
- Tertiary treatment to remove nitrogen and phosphorus – nitrogen usually removed by nitrification followed by denitrification,
- Phosphorous is removed by microbial uptake or chemical precipitation.

Biosorption and Bioaccumulation

- Biosorption is a very simple **physicochemical process** resembling conventional adsorption or ion exchange except that sorbent which in this case is the **material of biological origin**.
- The uptake of heavy metals (HMs) by microbial cells through biosorption is a metabolically passive process, which mostly occurs on the cells exterior.
- Biosorption can be carried out by **dead biomass** also as it occurs through ion exchange, surface complexation, and precipitation onto the cell wall and surface layers.
- **The bioaccumulation process is more complex than biosorption** itself and occurs in **two-stages**.
- This means that the first stage is biosorption and then in the second, they **become accumulated inside the cell** by - the most frequently - active transport system.
- The process is **metabolism-dependent process** and **requires cultivation of an organism** in the presence of toxic metals.
- In bioaccumulation, selecting of organisms plays an important role.
- Although, it is usually **slower than biosorption**, it is possible to **achieve higher metals removal** via bioaccumulation because cells offer more binding sites both, on the surface and inside the cell.
- In addition, bioaccumulation is a **non-equilibrium process** because binding of metals keeps increasing due to continuous growth of the biomass

The comparison between biosorption and bioaccumulation process (Chojnacka, 2010).

Biosorption	Bioaccumulation
Passive process	Active process
Biomass is not alive	Biomass is alive
Metals are bound with cellular surface	Metals are bound with cellular surface and interior
Adsorption	Absorption
Reversible process	Partially reversible process
Nutrients are not required	Nutrients are required
Single-stage process	Double-stage process
The rate is quick	The rate is slow
Not controlled by metabolism	Controlled by metabolism
No danger of toxic effect	Danger of toxic effects caused by contaminants
No cellular growth	Cellular growth occurs
Intermediate equilibrium concentration of metal ions	Very low equilibrium concentration of metal ions



Phytoremediation

- Phytoremediation is getting more attention as a promising technology for the environmental cleanup option, using specialized plants for removing of toxic metals and site restoration.
- The remediation process using plants is distinct and only selective potential plants (either native or genetically modified) can be involved to remove HMs from soil, water and air.
- Different phytotechniques such as phyto-extraction, phyto-volatilization, phyto-stabilization, phyto-degradation etc. have been studied in details for decontaminating land and water bodies.
- In general, various parts of plants play significant role in phytoremediation such as uptake by root, bioaccumulation and translocating them to the aerial parts through the designated tissues, transformation and/or degradation within the various plant parts. Finally, after harvesting the plant on maturity, the biomass is processed by drying, ashing or composting or can be reprocessed for further use.
- The **best plants which are used for phytoremediation should have the following traits:**
 - High biomass production,
 - A fast growth rate,
 - Extensive fibrous root systems,
 - Easy to cultivate and harvest,
 - Highly tolerant to contaminants, diseases and pests with an ability to accumulate high levels of contaminants

There are **several advantages** of phytoremediation, which are:

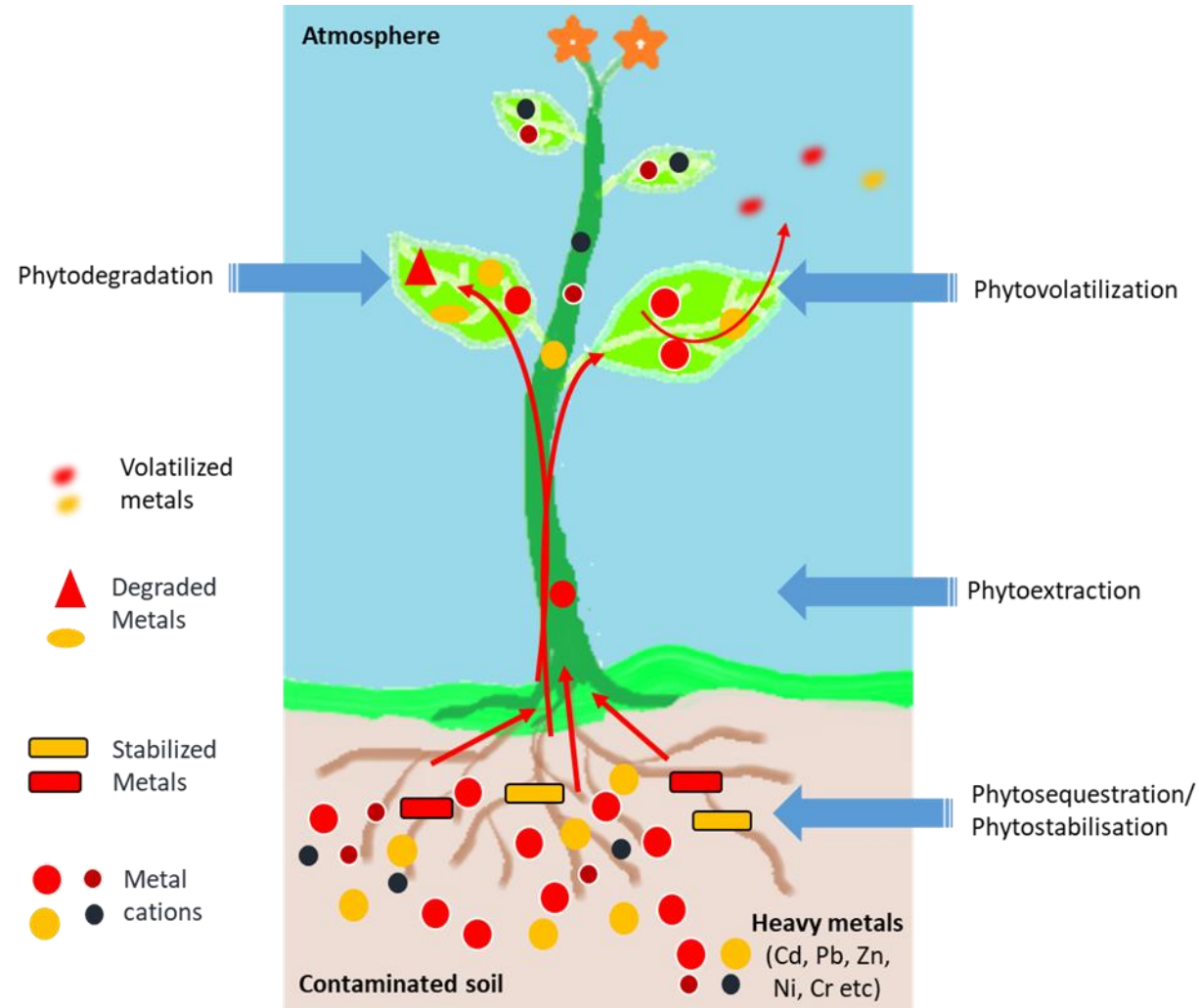
- It is an aesthetically pleasing, and publicly accepted and environmentally-friendly technology,
- Relatively cheaper,
- Can be applied against a wide range of contaminants
- Minimum environmental disruption, moreover it also reduces soil erosion due to plantation
- Produce recyclable plant residue with high metal content with a possibility of recovering and reusing of metals. The contaminated and unusable plant biomass can also be reduced to only 20 to 30 tons ash per 5000 tons soil by thermal or microbial or chemical process

Phytoremediation also has a **number of limitations**, which are:

- Phytoremediation is most effective only in the contaminated soil or the groundwater (< 5 m depth) due to its shorter roots.
- Takes much longer time clean up a site,
- Climate or seasonal conditions, organic and inorganic contaminants may influence the growth and survival of the plant. Thus proper care of the site is needed,
- Plants absorbing toxic HMs may pose significant threat of transferring HMs to the food chain and results in non-edible plant products
- Incomplete removal of HMs from the contaminated site. Moreover, volatilization of HMs may just convert a soil or groundwater contamination to an air contamination instead of complete bioremediation of HMs

Mechanisms of phytoremediation

- Phytoremediation generally happens via various mechanisms including phyto-sequestration, phyto-extraction, phyto-stabilization, phyto-degradation, phyto-volatilization and rhizofiltration.
- In fact, the five main subgroups involved in this processes function simultaneously, to some extent to remove the HMs from the contaminated sites.



Phytostabilisation and Phytosequestration

- Phytostabilisation/phytoimmobilization is a process where the toxic compounds immobilized or stabilized at contaminated sites thereby reducing the bioavailability of pollutants in the environment.
- Immobilization of HMs using certain plant species can be achieved through **metal sorption and accumulation** by roots through root hairs, precipitation, complexation or reduction of metal within the rhizosphere.
- However, the use of this strategy may have **notable limitations such as contaminant remaining in the soil, possible certain negative impacts of application of soil amendments for the stabilization of the contaminants and constant monitoring of the process**

Phytoextraction

- Phytoextraction is a common process used for removing toxic metals where plants absorb HMs from the contaminated site into the roots and **translocate them to the aboveground shoots or leaves.**
- This method is highly effective in removing metals/metalloids from polluted soils, sediments, water and **commercially feasible as well** compared to other techniques of phytoremediation.

Phytodegradation

- Phytodegradation/Phytotransformation is the process where contaminants are taken up by the plants **with the subsequent breakdown, mineralization, or metabolization to simpler less toxic forms.**
- Breakdown of the contaminants occurs by the plant itself through metabolic processes and various internal enzymatic reactions inside the plant.
- **Since HMs are nonbiodegradable, this process of phytoremediation is mainly limited to removal of organic impurities.**
- Some of the enzymes which help in the phytodegradation process are:
 - Dehalogenase - transformation of chlorinated compounds
 - Oxygenase - conversion of hydrocarbons such as aliphatic and aromatic compounds,

Phytovolatilization

- This process involves the uptake of toxic metals by root system, translocation to shoot system through conducting tissues and finally release/volatilization of metals in the less toxic form into the atmosphere from the stem/trunk and leaf surfaces.
- It can especially be applied for the remediation of volatile organic compounds (VOCs).
- Similarly, certain toxic metals like Hg, Se can be uptaken from the contaminated sites and biomethylated to form volatile compounds such as mercuric oxide and dimethyl selenide, which further evaporated or volatilized into the atmosphere
- For example, mercuric ion can be transformed into the less toxic gaseous elemental Hg by plants, but there are no or very limited naturally occurring plants that can perform this.